

CineVerse- Movie Ticket Booking System

Group No 05

Course: Object-Oriented Programming with Java

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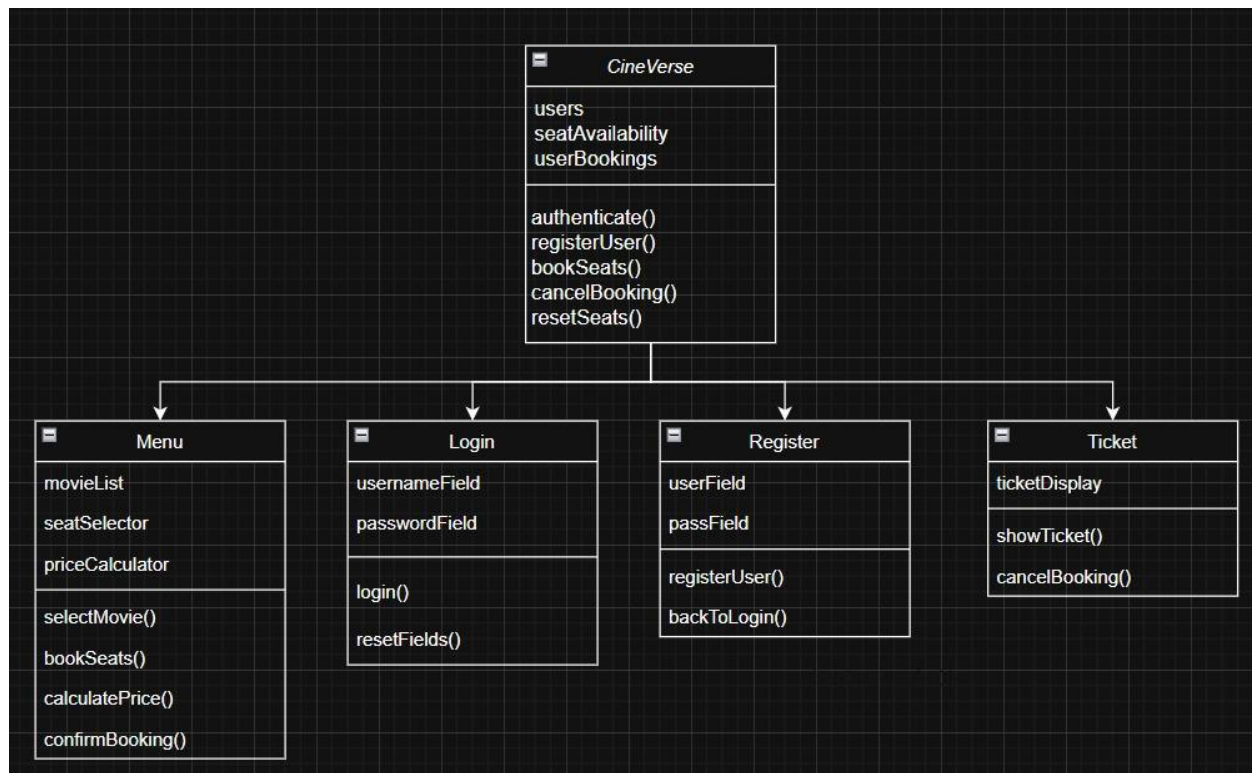
Real world problem Description

In today's fast world, people love watching movies but often face problems when buying tickets. They have to wait in long queues, and sometimes seats are sold out before they reach the counter. It also becomes difficult to manage seat availability and bookings manually.

To solve this problem, Cineverse provides an easy way to book movie tickets through a simple computer-based system.

It allows user to register, log in, choose their favorite movie, select show time, and seat type (regular or VIP), and confirm bookings easily. It also includes features like seat availability checking, cancellation and secure admin reset of all seats. This makes movie ticket booking faster, more convenient and user friendly.

Class Diagram



How to OOP Concepts are Applied ?

Encapsulation	All important data like users,seat availability ,and bookings are stored inside private variables (e.g , users,seatAvailability) within the cineverse class. Public methods (like registerUser() ,bookSeats(), and cancel booking()) are used to access and modify these safely.
Abstraction	The user doesn't see how the booking logic or seat handling works internally.They only interact with buttons and forms on the GUI - which simplifies the complex logic behind ticket management .
Inheritance	While direct inheritance between classes isn't used,all panes (LoginPanel,RegistrationPanel, MenuPanel,TicketPanel) inherit from the JPanel class - a built in Swing class to reuse GUI functionality
Polymorphism	Event listeners and overridden methods (like actionPerformed()) demonstrate polymorphism,where the same action interface behaves differently for login ,registration ,and booking operations.
Composition	The main class Cineverse creates and contains multiple GUI panels. These panels work together as part of the main program,showing a real world example of composition .

CineVerse - Movie Ticket Booking

CINEVERSE LOGIN

Username:

Password:

Login

Reset

Reset All Seats

Register

CineVerse - Movie Ticket Booking

REGISTER ACCOUNT

Choose Username:

Choose Password:

Register

Back to Login

CineVerse - Movie Ticket Booking

Select Movie Details

Movie:

Avengers

Show Time:

10:00 AM

Seat Type:

Regular

Seat:

B1

Quantity:

-

1

+

Calculate Price

Per seat: \$15.00 | Total: \$15.00

Availability:

10 seats available

Logout

Confirm Booking

Screenshots of GUI

CineVerse - Movie Ticket Booking

Your Ticket

CINEVERSE TICKET

User: seewini

Movie: Avengers

Time: 10:00 AM

Seat: B1

Type: Regular

Quantity: 1

Per Seat: \$15.00

Total: \$15.00

Enjoy your movie! 🍿

Back to Menu

Cancel Booking

Logout

Your Ticket

```
-----
CINEVERSE TICKET
-----
User: seewini
Movie: Avengers
Time: 10:00 AM
Seat: B1
Type: Regular
Quantity: 1
Per Seat: $15.00
Total: $15.00
-----
Enjoy your movie! 🍿
```

Cancel Your Booking

×

Seat:

B1

▼

Quantity to cancel:

1

+

-

OK

Cancel

Exception handling

The system uses pop up dialogs (JOptionPane) to handle and display errors such as

- Invalid username and password
- Empty field during registration
- Trying to book unavailable seats
- Entering wrong admin password during seat reset

This ensures the program does not crash and gives friendly messages to users.

Event handling

All actions (like login,register calculate price, confirm booking ,cancel booking,etc) use ActionListeners.

```
btnLogin.addActionListener(e -> {  
    String user = txtLoginUser.getText();  
    String pass = new String(txtLoginPass.getPassword());  
    if (parent.authenticate(user, pass)) {  
        parent.showCard("menu");  
    } else {  
        JOptionPane.showMessageDialog(this, "Invalid username or password!");  
    }  
});
```